Budget

Amazon Research Awards, Fall 2026 — Amazon Security
Measuring Feedback-Driven Improvement in Lightweight Code-Security Verifiers for Spec-Driven Code Generation

PI: Younghun Chae, Associate Professor, Department of Computer Science, Kent State University at Stark — SOLO, no Co-PI

Cash funding needed: $80,000 USD
AWS Promotional Credits needed: up to $40,000 USD (conditional, stretch tier)

Funds Needed

We request $80,000 in cash funding to support the full 12-month execution of this project, and, conditional on award, up to $40,000 in AWS Promotional Credits as a stretch tier. Required scale evidence for this proposal is local-capacity only (verifier latency/throughput and loop-iteration cost on local GPUs); the AWS credits validate the same pipeline at larger multi-instance scale and gate no core deliverable. All requested funds are applied to direct project expenses; no institutional overhead is requested.

A. Cash Funding ($80,000)

- **Graduate Student, hourly ($30,000)**: 1,500 hours at $20/hour, the rate this lab already uses for graduate research assistants on an active sponsored project. This individual leads the oracle pipeline implementation, verifier fine-tuning, and the feedback-loop workload described in the Methods section.
- **Undergraduate Student, hourly ($12,000)**: 800 hours at $15/hour, a standard Kent State undergraduate research-assistant rate (pending institutional confirmation). Covers corpus curation, harness maintenance, and evaluation support under graduate-student and PI supervision.
- **PI effort ($25,000)**: Summer-rate-equivalent continuous effort over the 12-month project, increased relative to a single-student budget to reflect direct supervision of a two-tier (graduate plus undergraduate) student team; covers system design, oracle-protocol specification, and dissemination.
- **Domestic Travel ($5,000)**: Supports travel to two conferences (e.g., IEEE S&P, USENIX Security, or ISSTA) to present interim and final (M12) results and engage with the research community.
- **Equipment / storage / oracle infra ($4,000)**: Sandboxes, CPU fuzz instances, and CodeQL/Semgrep compute for the oracle battery; institutional (non-AWS) compute that complements the AWS credits request.
- **External inference ($3,000)**: Frontier LLM judge baseline (API) used for the cost-quality Pareto comparison in the Evaluation section; estimate, revisited at the M1 pilot cost measurement.
- **Contingency ($1,000)**: Unallocated buffer for unforeseen direct costs.

All cash funding ensures full coverage of research personnel, travel, software/infrastructure, and the external-inference comparison arm, while maintaining compliance with ARA's "direct cost only" policy. Single-institution (Kent) solo budget: one graduate student, one undergraduate student, and PI effort; no Co-PI, no sub-award. Student cost is hourly wages only — tuition, fringe, and fully-loaded treatment are pending institutional confirmation and excluded from the cash total above.

B. AWS Promotional Credits (up to $40,000, conditional stretch tier)

To validate the verifier and feedback loop at AWS production scale beyond the local-GPU pilot, we request AWS credits allocated as follows:

- **Amazon SageMaker ($20,000)**: Multi-arm LoRA fine-tuning and evaluation of the verifier at scale, covering the main feedback-loop and ablation-arm training workload.
- **Amazon EC2, compute-optimized ($7,000)**: CPU-bound fuzz/eval farm (Atheris / Jazzer.js harnesses) for the oracle battery at scale.
- **Amazon EC2, g-series (GPU) ($5,000)**: GPU-bound inference and evaluation of the verifier at scale, complementing the SageMaker training allocation.
- **Amazon S3 + data transfer ($4,000)**: Corpus, traces, and artifact storage, plus inter-service data transfer.
- **AWS CloudWatch ($2,000)**: Pipeline monitoring and run-level logging across the scaled oracle and training jobs.
- **Buffer ($2,000)**: Reserved for instance overage and S3 egress as experimental scale is tuned.

AWS credits are essential for validating the verifier and feedback loop under realistic production-scale conditions; they are stretch-only, since every primary deliverable (corpus, verifier, loop, escalation benchmark) is credit-independent and locally reproducible on university infrastructure. AWS Public Dataset usage: N/A — the corpus is self-generated (public-repository task instances plus CVE-sourced and seeded-bug instances built by this project), so there is no public-dataset dependency. The allocations above are estimated shares, not unit prices; M1 fixes core/stretch experiment counts and per-product unit-cost assumptions, revisited if realized costs diverge, consistent with the Proposal document.